

Program Function Introduction

We wrote three core scripts in the Unity scene:

1. ImageCapture.cs Mounted on the left and right cameras.

Function: Save the camera's rendered images as PNG pictures; support sending images to MATLAB in real-time via TCP protocol.

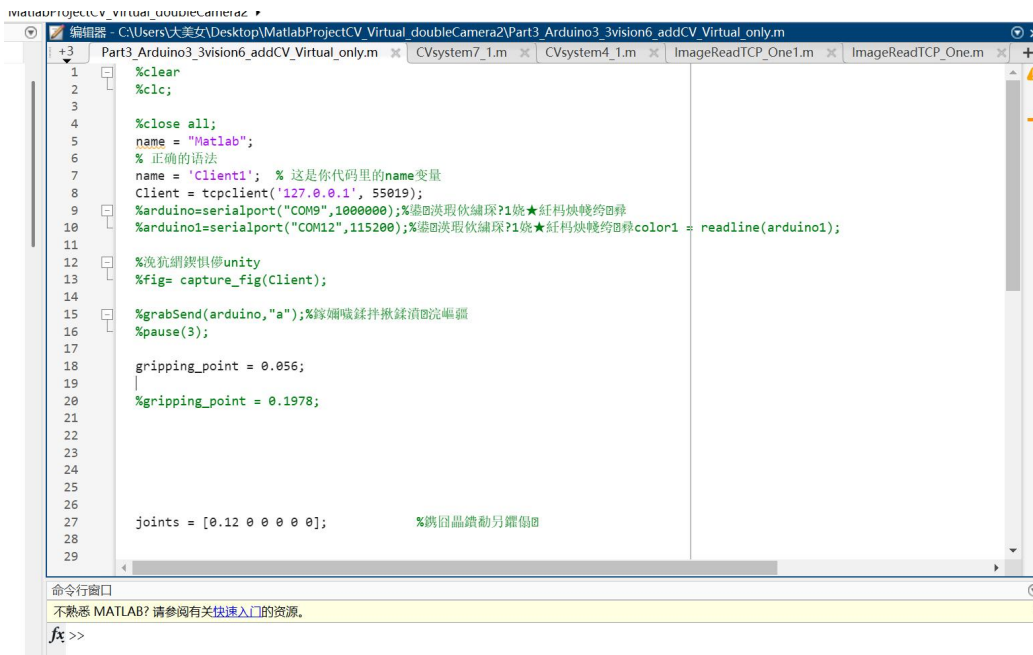
2. LaserController.cs Control the switch and rotation angle of the laser emitter

Function: Simulates real-line structured light projection, can dynamically adjust the orientation of the laser plane, and generate different light stripe patterns.

3. UIManager.cs Handle UI button click events

Function: Respond to operations such as 'Capture calibration image', 'Start monocular localization', 'Start binocular localization', 'Switch target position/shape', 'Reset scene', etc., and call the corresponding image acquisition or data transmission functions.

MATLAB-side program



```
1 %clear
2 %clc;
3
4 %close all;
5 name = "Matlab";
6 % 正确的语法
7 name = 'Client1'; % 这是你代码里的name变量
8 Client = tcpclient('127.0.0.1', 55019);
9 %arduino=serialport("COM9",1000000);%通过串口发送数据?1姚★红杓映映约@群
10 %arduino1=serialport("COM12",115200);%通过串口发送数据?1姚★红杓映映约@群color1 = readline(arduino1);
11
12 %通过串口接收数据
13 %fig= capture_fig(Client);
14
15 %grabSend(arduino,"a");%通过串口发送数据?1姚★红杓映映约@群
16 %pause(3);
17
18 gripping_point = 0.056;
19 |
20 %gripping_point = 0.1978;
21
22
23
24
25
26
27 joints = [0.12 0 0 0 0 0]; %通过串口发送数据?1姚★红杓映映约@群
28
29
```

1. Monocular positioning program mono_localization.m

Function: Using images from a single fisheye camera, combined with the laser plane equation, to calculate the three-dimensional coordinates of the target.

Internally calls the following sub-functions:

extract_laser_center.m: Sub-pixel laser stripe center extraction based on the Hessian matrix (Steger algorithm).

cluster_points.m: DBSCAN clustering for denoising, retaining the cluster with the most points, and calculating the average coordinates.

2. Binocular positioning program binocular_localization.m

Function: Uses images from the left and right fisheye cameras to directly calculate three-dimensional coordinates through triangulation, without relying on the laser plane equation.

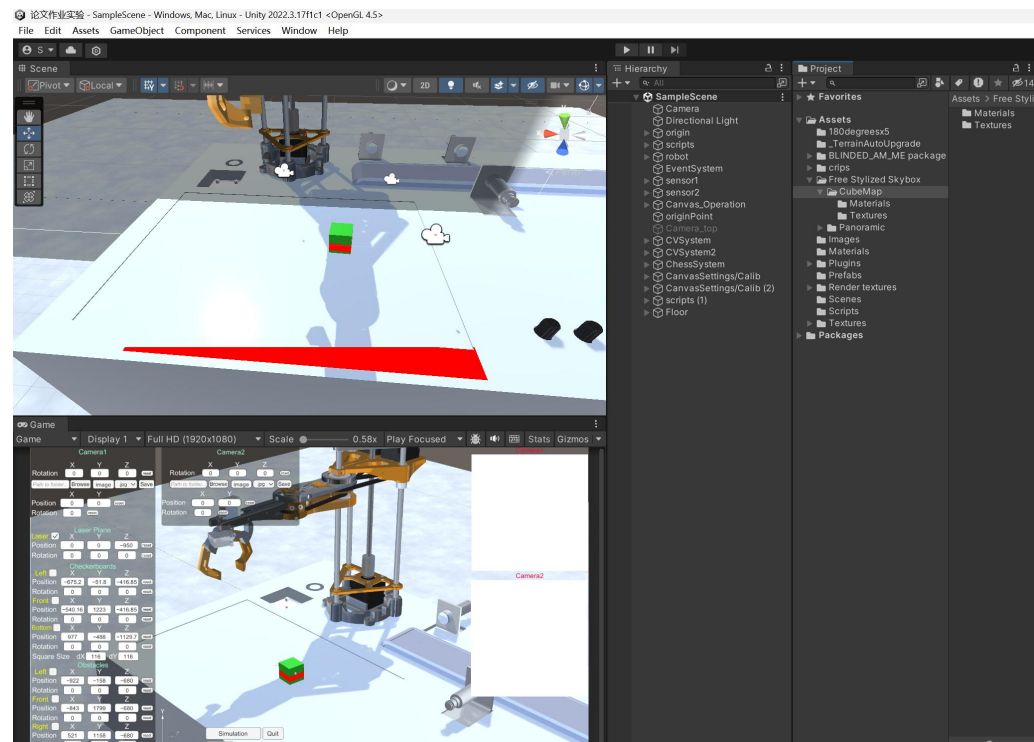
Internal calls:

extract_laser_center.m (extracts the center of the light stripe on the left and right images, respectively).

triangulate.m: Performs triangulation on matching points according to the binocular external parameters to obtain spatial points.

cluster_points.m: After merging the left and right point clouds, clusters and denoises them, outputting the average coordinates of the target.

How to use the Unity3D virtual simulation platform we created in combination with MATLAB image processing and positioning algorithms to achieve high-precision target positioning based on multi-view fisheye cameras and linear structured lighting.

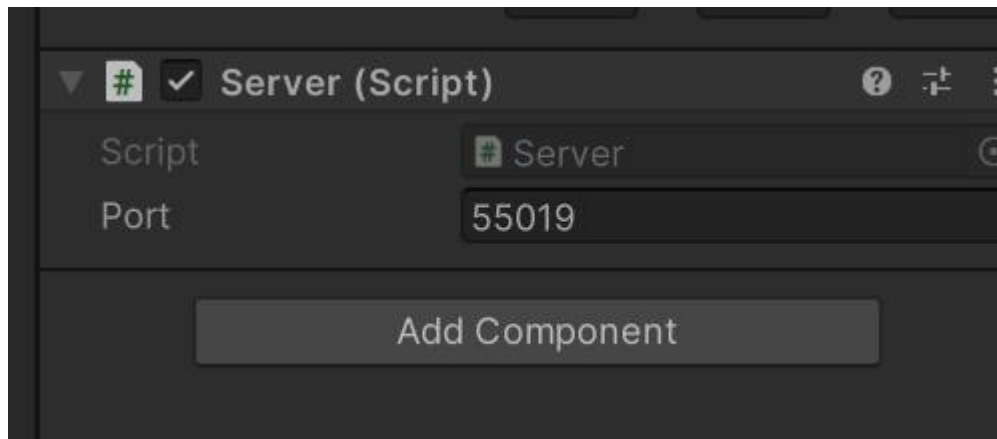


Сначала откройте симуляционную сцену Unity. Наша система включает две 180° рыбий глаз

камеры, названные левая камера CVSystem и правая камера CVSystem2; проводной лазерный излучатель Sensor; а также различные объекты для тестирования, такие как здесь куб, сфера и так далее.

It is necessary to ensure that the Unity script matches Matlab and equals 55019

```
name = 'Client1'; % 这是你代码里的name变量
Client = tcpclient('127.0.0.1', 55019);
```



% 同 鳳 湖 帆 月 測 鋒 劍 夢 道 嗟 匪 史 恪 惛 詞 慳 况 沉 咽 猶

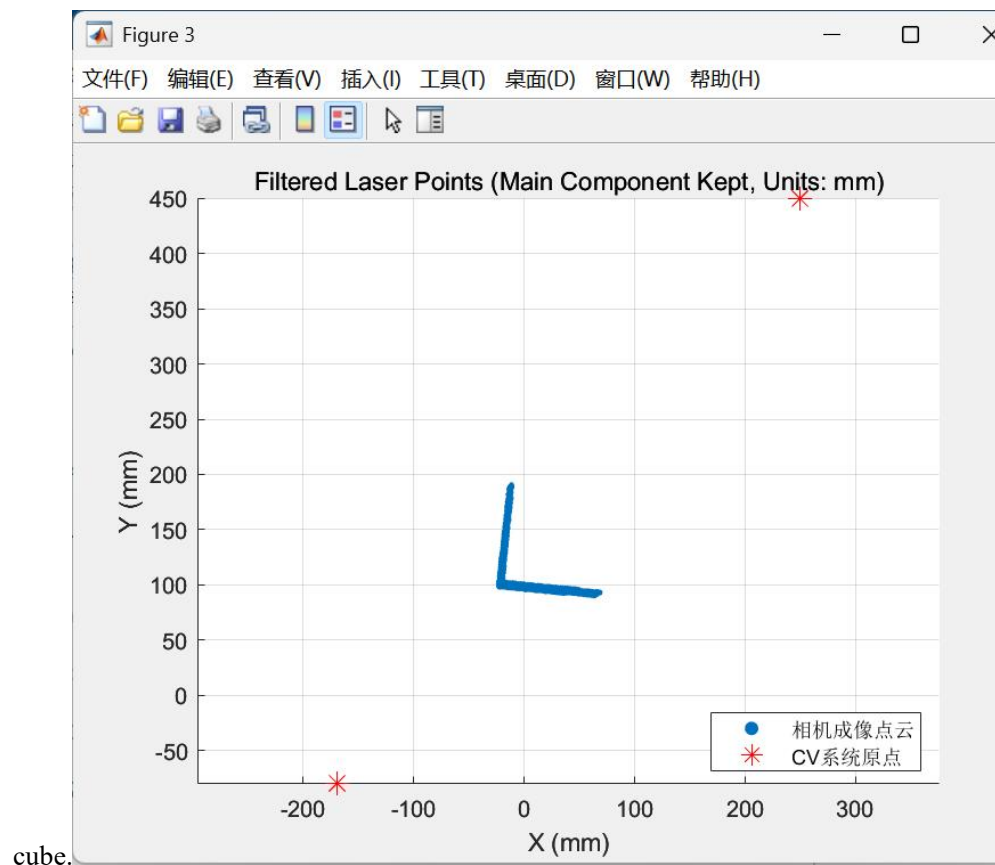
```
Loc=CVsystem7_1(Client);%CVsystem 寡 冲 澈 鍊 肩 睦 娉 囉 紅 CVsystem2 鑒 欬 漠 灝 忤 ? 肩 睦 娉 ?, C
pot=PositionTran(Loc(1),Loc(2));
t = 0:0.1:6;
X1 = pot(1,1)/100; % Z
Y1 = pot(2,1)/100; % -X
Z1 = -0.01; % Y - GREEN
```

This position can switch, but the binocular scheme

Currently, a stereoscopic scheme is being executed, the cube is moving and Unity is starting, after that the main MATLAB program is launched

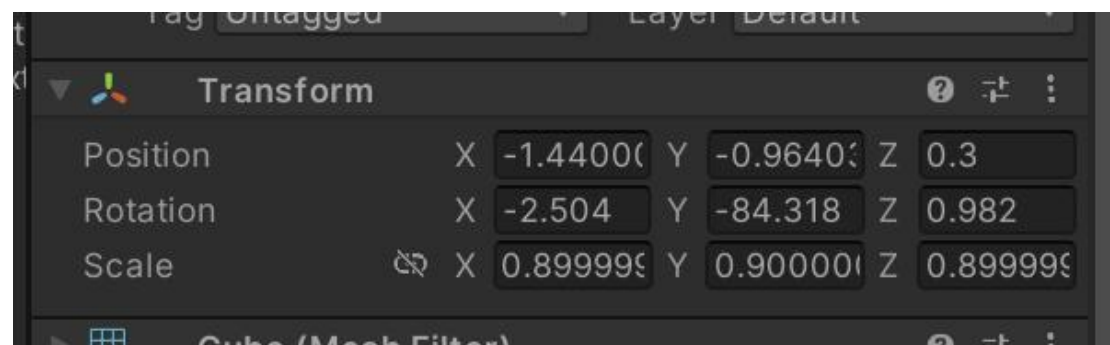
Next, an experiment on monocular localization is conducted. As an example, let's take a cube and place it in the first test position. The laser plane is projected onto the surface of the cube, forming a bright line of light. The left camera captures this image and transmits it to MATLAB in real time via TCP.

The MATLAB program first precisely extracts the coordinates of the center of the light line with sub-pixel accuracy using the Hessian method, then, using the pre-calibrated internal camera parameters and the laser plane equation, calculates the three-dimensional coordinates of each point on the light line. Since the image may contain background noise or reflected interference, we use clustering based on Euclidean distance, keeping only the cluster with the largest number of points, after which we average it as the final position of the



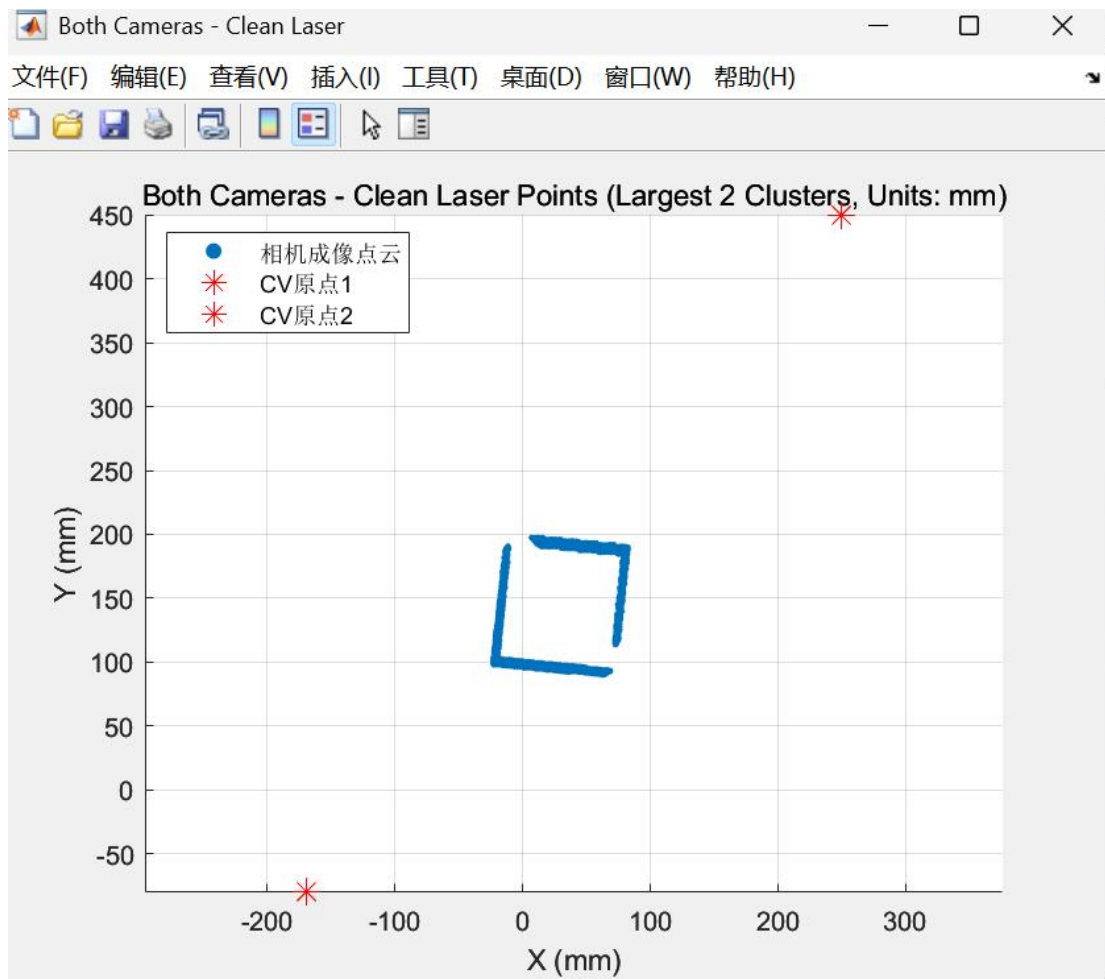
$$x = 4.61, y = 112.59$$

Это исходное положение цели в Unity



Switch to stereo camera mode to show simultaneous shooting from the left and right cameras, run the MATLAB stereo script

```
% 閩氫繡楸肩濼鋒勸儇澶磋 纒格柵錡榑殞浣
Loc=CVsystem4_1(Client);%CVsystem 奪
pot=PositionTran(Loc(1),Loc(2));
```



$$x = 24.08, y = 137.68$$

The binocular scheme can further improve accuracy. Here we simultaneously use two fisheye cameras — the left and the right for shooting, after which MATLAB reads the two images and separately extracts the centers of the light stripes. The key difference is that the monocular system must rely on the calibration values of the laser plane, whereas the binocular system directly uses the disparity of the images for triangulation and is not sensitive to errors in the laser plane.

The program merges point clouds from two viewpoints, then performs clustering to remove noise, and the coordinate accuracy calculated by the binocular system at the same location is significantly higher than that of the monocular system.